Venkata Lakshmi Parimala Pasupuleti

Software Engineer · Python · React · TypeScript · PostgreSQL · HUD & Affordable Housing Compliance · Property Management SaaS · 9yr Production

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github.com/venkatalakshmiparimala | Virginia, USA (open to Atlanta, GA)

PROFESSIONAL SUMMARY

Software engineer with 9+ years of production experience building, enhancing, and maintaining the same HUD affordable housing compliance systems Yardi's HUD Business Unit develops - at Entrata, Yardi's direct competitor. For 7 years I owned the HUD federal compliance pipeline end-to-end: annual income limit imports (HUD federal source data to 180,000 domain records), rent, accounting, and eligibility calculations for 100K+ housing units, RabbitMQ-based async processing, and AWS S3 document storage - the same affordable housing data workflows the Yardi HUD team ships and maintains. I debug with confidence: 200+ SQL optimizations (execution plan analysis, indexing, partitioning), internal debugging tools that cut triage time 40%, and bug count maintained below 10 for 6+ consecutive months. I ship full-stack: Python and PHP backends, PostgreSQL/MySQL/Oracle databases, React.js and TypeScript frontends, REST APIs, and AWS infrastructure - 20+ production modules across 120+ Agile sprints at 99.5% uptime. I also build AI systems (LeaseI: 8-agent LangGraph compliance platform; AgentProbe: LLM evaluation harness) - but the core of my work is reliable, well-tested software that property management teams can depend on every day. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

Languages & Fullstack: Python (9yr), PHP (2yr production SaaS), TypeScript, JavaScript, React.js, SQL, PL/SQL, Bash, HTML/CSS, jQuery, FastAPI, Flask, REST APIs, Linux (production 9yr), Git, Agile/Scrum

Databases & Infrastructure: PostgreSQL, MySQL, Oracle, SQL Server, AWS (S3, EC2, RDS, Glue), Docker, Microservices, RabbitMQ (event-driven async, retry, idempotency), Redis, GitHub Actions CI/CD, Automated Testing, Performance Engineering, Query Optimization, Execution Plan Analysis, Indexing, Partitioning

Domain & Compliance: HUD Federal Compliance, Affordable Housing (LIHTC, Section 8, Section 42), Income Limit Imports, Rent Calculations, Eligibility Workflows, PHI/HIPAA Compliance, Audit Logging, RBAC, Data Governance, Regulatory Compliance Pipelines, Property Management SaaS, LangGraph, LangChain, AI/ML Integration

PROJECTS

LeaseI: HUD Compliance AI System - Property Management Domain

HUD · Compliance · Audit Logging · Property Management · Production

Python · LangGraph · FastAPI · OCR · SQLite · REST APIs · Claude Code | github.com/venkatalakshmiparimala/Lease_complaince_rag

Production compliance system for affordable housing lease and regulatory workflows - built on deep knowledge of HUD federal rules, income limits, and eligibility requirements. **Domain:** processes lease documents, HUD regulatory data, and compliance decisions across 1,240+ test cases with 93.6% precision. **Architecture:** 8-agent LangGraph pipeline with conditional routing, shared SQLite knowledge store, and deterministic HITL guardrails ensuring no auto-resolution on regulated decision paths. **Auditability:** complete decision trail with timestamp, source, confidence, and outcome for every action. CI regression gate blocks quality drift before deployment.

AgentProbe: Production Evaluation Platform - Observability & Quality

Observability · REST APIs · Testing · Open Source

Python · FastAPI · LangGraph · ChromaDB · GitHub Actions · Anthropic API · OpenAI API · Ollama | github.com/venkatalakshmiparimala/AgentProbe

Scalable production platform with clean API design and high-quality engineering standards. **Backend:** 15-endpoint FastAPI REST API with multi-provider routing, clean abstractions, and secrets management. **Quality:** 5-metric observability engine; CI-integrated regression gate blocking drift before deployment; zero regressions across 1,240+ test cases. **Delivery:** shipped end-to-end as sole engineer - requirements, design, implementation, testing, and open-source release. Enterprise-deployable.

AI on FHIR: Fullstack Application - REST APIs, RBAC & Data Isolation

Fullstack · REST APIs · HIPAA · React · TypeScript

Python · Flask · React.js · TypeScript · Docker · BERT · spaCy · FHIR REST APIs | github.com/venkatalakshmiparimala/AI_on_FHIR

HIPAA-compliant fullstack application - Python/Flask backend, React.js + TypeScript frontend, Docker containerization; RBAC with authentication and full data isolation; complete audit trail; ~60% reduction in manual retrieval time. Demonstrates the secure fullstack integration pattern property management platforms implement for resident data, document storage, and compliance reporting.

EXPERIENCE

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer - Production Compliance System Delivery

- Designed and delivered a production compliance AI system (Python, LangGraph, FastAPI) end-to-end for an external customer - requirements, architecture, implementation with audit logging and human-review gates, automated testing, and production deployment in weeks as sole technical owner.

George Mason University

Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Built ML pipelines (PyTorch, scikit-learn, R^2=0.977) on WHO/UK datasets; designed observability dashboards improving research comprehension 40%; mentored 600+ students in Python, SQL, and software engineering.

Entrata - Apartment & Affordable Housing SaaS (Yardi Competitor)

Sep 2018 - Aug 2024

Senior Software Engineer · Lehi, UT (Remote)

- **HUD federal compliance pipeline ownership (7 years)** - sole owner of the annual income limit import pipeline (HUD federal source data to 180,000 domain records driving rent calculations and eligibility decisions for 100K+ housing units); RabbitMQ event-driven async processing with retry logic and idempotency; AWS S3 document migration (100% PHI compliance, zero data loss); 99.5% uptime across the full ownership period.
- **Full-stack development and bug resolution** - 20+ full-stack production modules (PHP backend, Python data pipelines, PostgreSQL/MySQL/Oracle, JavaScript/TypeScript, React.js); 200+ SQL optimizations (30% performance gain via execution plan analysis, indexing, partitioning); internal debugging tools reducing triage time 40%; bug count maintained below 10 for 6+ consecutive months; GitHub Actions CI/CD with automated testing throughout.
- **Technical collaboration and delivery** - 120+ Agile sprints with product, compliance, and operations stakeholders; presented implementation details and solutions to internal teams; 300+ peer code reviews; mentored 5+ engineers on production practices and code quality; Python/Bash automation eliminating 80% of manual workload.

Entrata India - Pune

Jan 2017 - Sep 2018

Associate Software Engineer → Trainee

- Built backend modules (PHP, Python, PostgreSQL) for production affordable housing SaaS; HUD compliance data pipeline for 1.6M records across 5 program years; AWS S3 migration resolving 90% of legacy storage constraints.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017